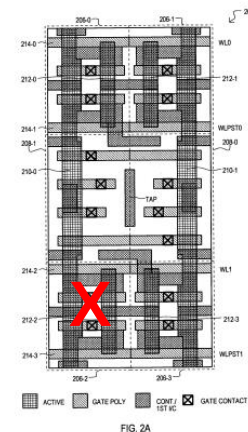


Fault Modeling

- Introduction
- **Fault Models**
 - ◆ Stuck-at fault (1961)
 - ◆ Bridging fault (1973)
 - ◆ Delay fault (1974)
 - ◆ Transistor-level fault
 - * Stuck-open (1978)
 - * Stuck-on
 - * Cell-aware (2009)
 - ◆ Review
- Fault Detection
- Fault Coverage
- Conclusion



Why Faulty NAND Has Memory ?

- Stanford Murphy Experiment [Li 01]

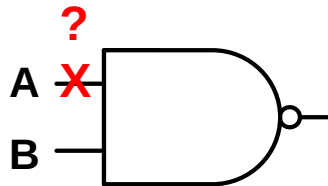
- ♦ **Sequence Dependency:** Test result depends on test sequence

- Example: NAND gate

- ♦ Same test patterns {11, 01, 00}, but in different sequence

- ♦ Sequence 1 **fails**; Sequence 2 **passes**

- ♦ Is it **A SA1** or not?



Sequence 1

A	B	Out _{good}	Out _{faulty}
1	1	0	0
0	1	1	0
0	0	1	1

← A SA1 ?

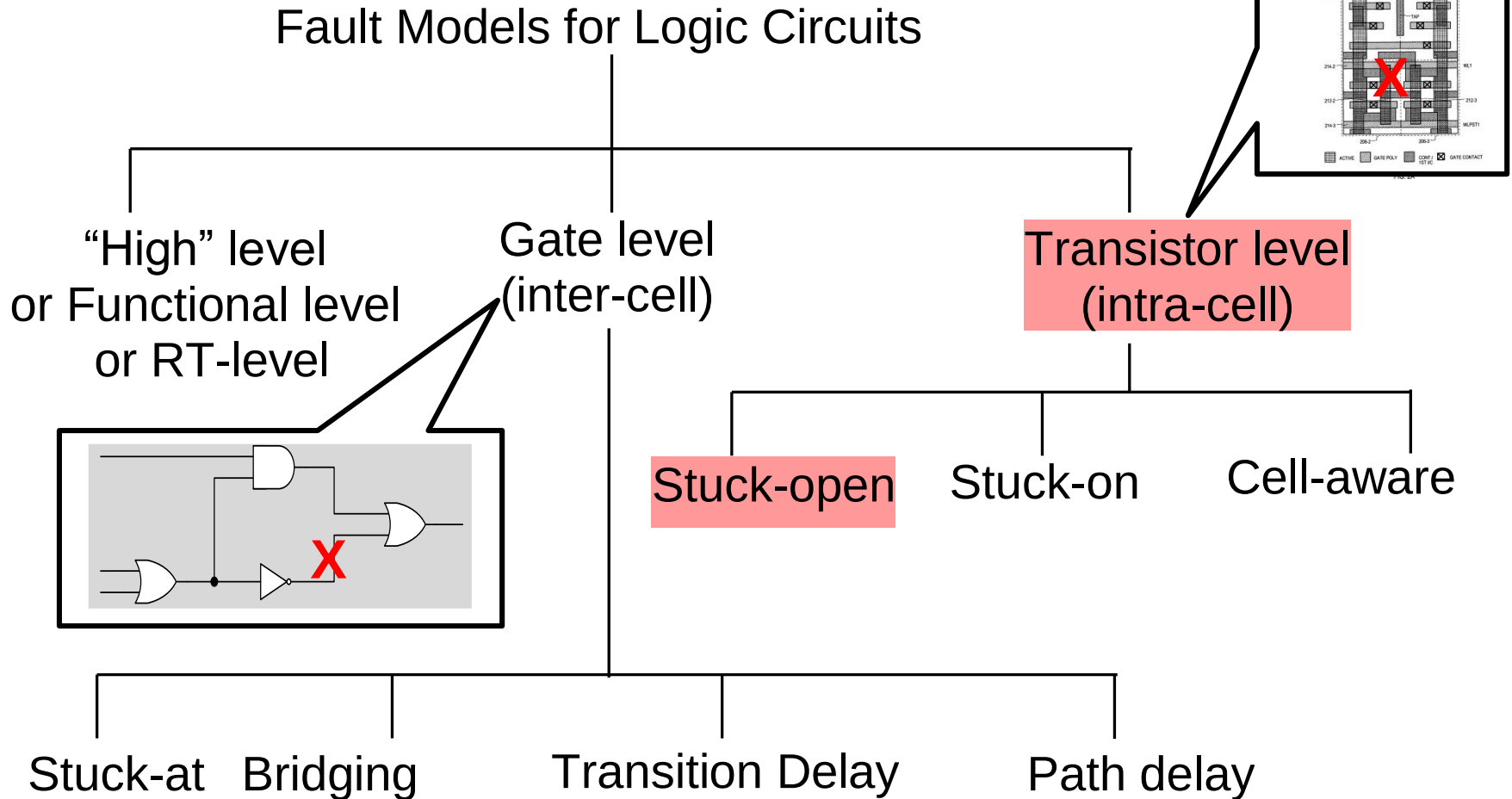
Sequence 2

A	B	Out _{good}	Out _{faulty}
0	0	1	1
0	1	1	1
1	1	0	0

← Not A SA1 ??

**SSF Cannot
Explain Seq. Dep.**

Classification of Fault Models



Stuck-open Fault [Wadsack 78]

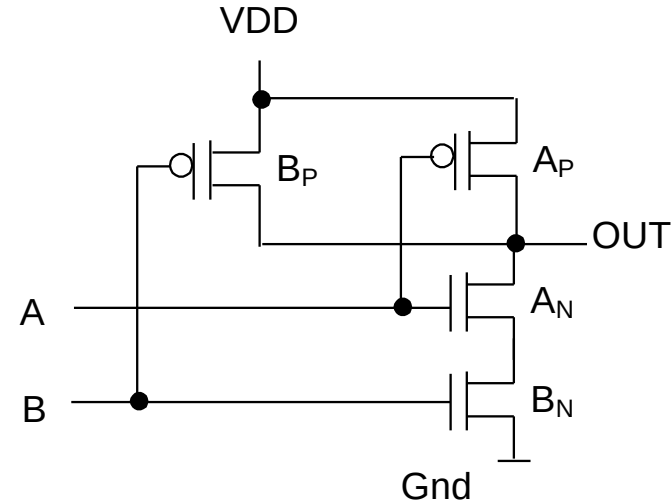
- **Stuck-open (SOP) fault**

- ♦ Faulty transistors is always off

- **Example: NAND**

- ♦ **11→01** detects A_P SOP

- ♦ **Totally 4 faults: $A_P B_P A_N B_N$**

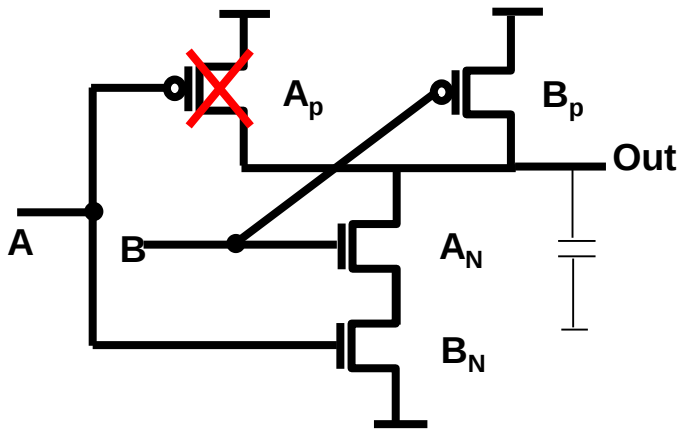


Input A B	Fault-free Output	Faulty Outputs Z = high impedance			
		A_P	A_N	B_P	B_N
0 0	1	1	1	1	1
0 1	1	Z	1	1	1
1 0	1	1	1	Z	1
1 1	0	0	Z	0	Z

SOP Requires 2-pattern Test

Sequence Dependency

- SOP Faulty circuit has *memory* (due to parasitic capacitor)
 - ♦ Test results depend on sequence of test patterns
- Example: A_p stuck-open fault,
 - ♦ Same test patterns{11, 01, 00}, but different sequence
 - ♦ Sequence 1 **fails**; Sequence 2 **passes** (test escape)



Sequence 1

A	B	Out _{good}	Out _{faulty}
1	1	0	0
0	1	1	0
0	0	1	1

FAIL:
Fault detected

Sequence 2

A	B	Out _{good}	Out _{faulty}
0	0	1	1
0	1	1	1
1	1	0	0

PASS:
Fault undetected
(test escape)

Simulation Results [Millman 89]

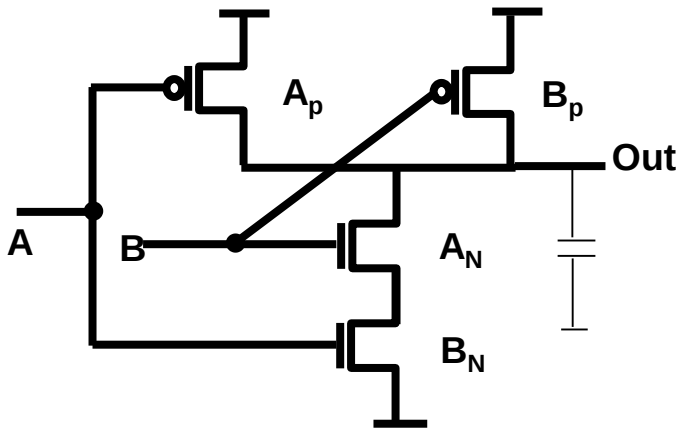
- Q: How effective is SSF test sets for stuck-open faults?
- 74LS181 ALU
 - ♦ Total **426** stuck-open faults
 - ♦ 100% SSF tests cannot achieve 100% SOP fault coverage

Test set	Krish	Br2	Goel	Mc1	Hugh
Test Length	12	14	35	124	135
SSF fault coverage %	100	100	100	100	100
SOP fault coverage %	79.3	79.6	90.8	87.1	93.4

SSF Not Enough for SOP

SOP $\neq$ TDF [Li 02]

- TDF is correlated with SOP fault, but they are NOT the same
- $\{11 \rightarrow 01, 01 \rightarrow 10, 00 \rightarrow 11\}$ has 100% TDF coverage
 - ♦ but only 75% SOP fault coverage (missed B_p)



$A_1B_1 \rightarrow A_2B_2$	Out_{good}	Detected SOP	Detected TDF
11 $\rightarrow$ 01	0 $\rightarrow$ 1	A_p	A STF; Out STR
10 $\rightarrow$ 01	1 $\rightarrow$ 1	-	A STF
11 $\rightarrow$ 10	0 $\rightarrow$ 1	B_p	B STF; Out STR
01 $\rightarrow$ 10	1 $\rightarrow$ 1	-	B STF
00 $\rightarrow$ 11	1 $\rightarrow$ 0	$A_N B_N$	A STR; B STR Out STF

TDF Not Enough for SOP, Either

Experimental Results [Woodhal 87] [Li 01]

- 1 μm CMOS [Woodhal 87]
 - ◆ Total population: 4,552 chips (passed parametric test)
 - * 1,225 chips failed stuck-at test
 - * 44 chips (3.6% of 1225) failed stuck-open test
 - * 4 chips failed stuck-open test **escaped** stuck-at test
- 0.7 μm CMOS, *Stanford Murphy experiment* [Li 01]
 - ◆ Total population 5.5K chips tested
 - * 116 defective chips
 - * 7 single stuck-open fault diagnosed
 - * 2 multiple faults (stuck-at + stuck-open)
- About 5% of failed chips are stuck-open

Stuck-open Cannot be Ignored

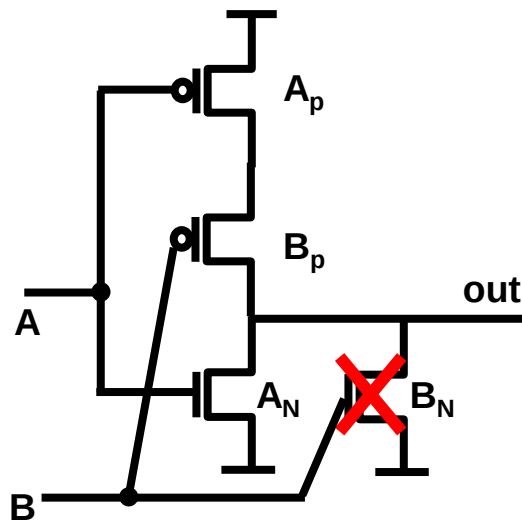
Quiz

Q1: Find a sequence of 2 patterns to detect B_N SOP fault in NOR

A:

Q2: Find a sequence of 2 patterns that cannot detect B_N SOP fault (test escape)

A:



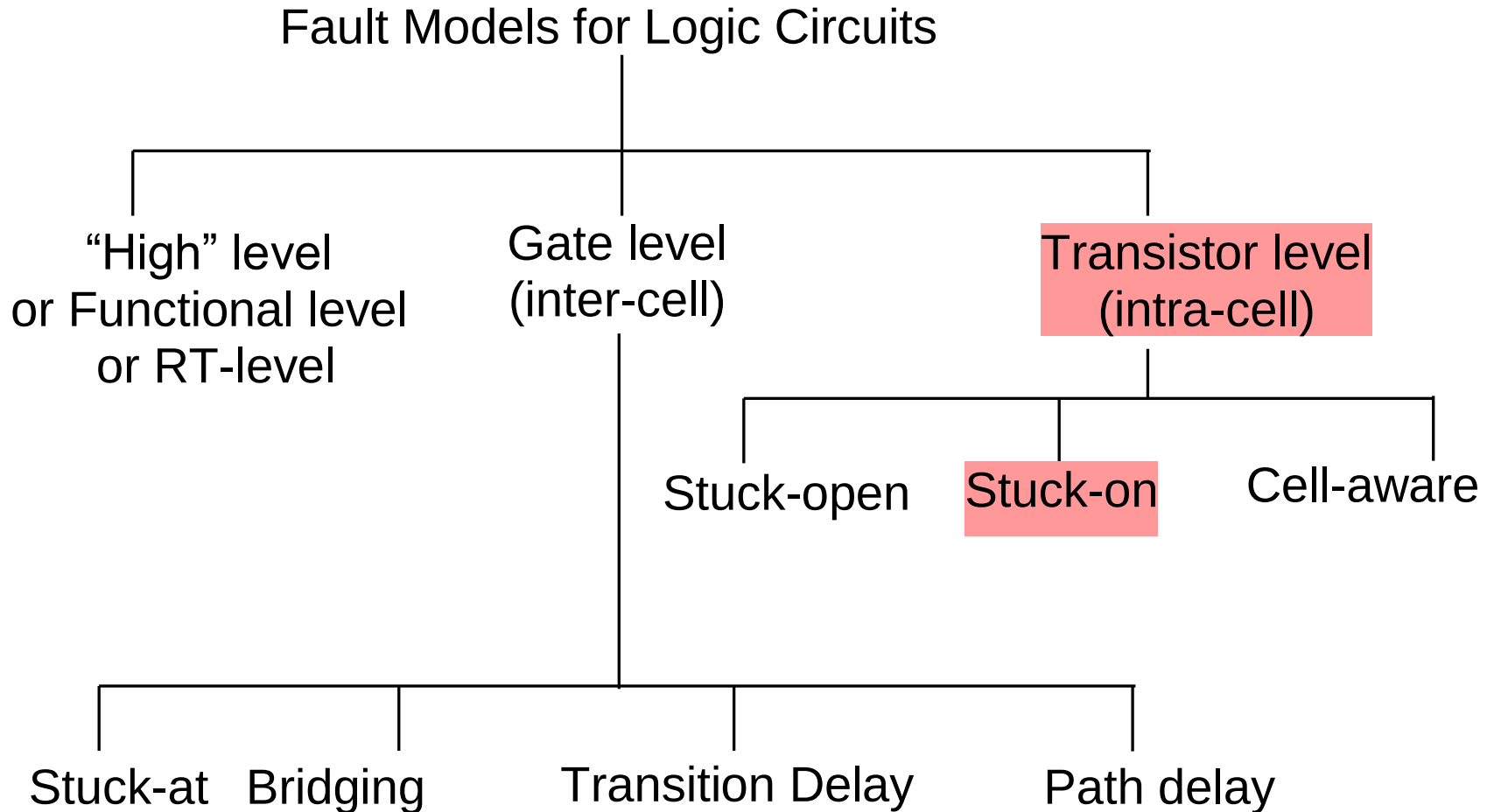
Sequence 1

A	B	Out _{good}	Out _{faulty}

Sequence 2

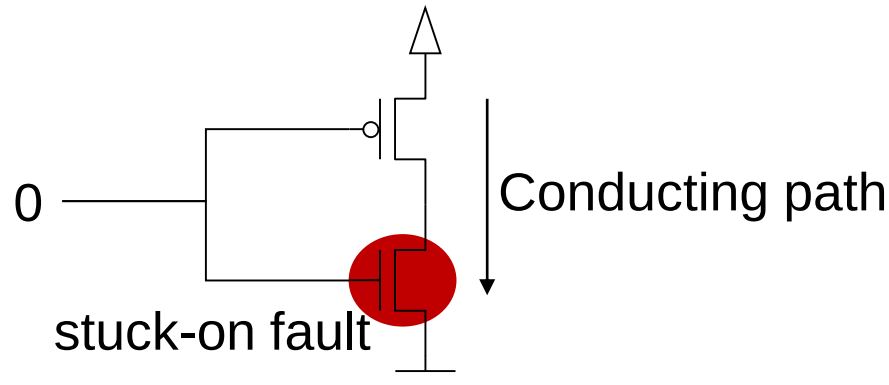
A	B	Out _{good}	Out _{faulty}

Classification of Fault Models



Stuck-On Fault

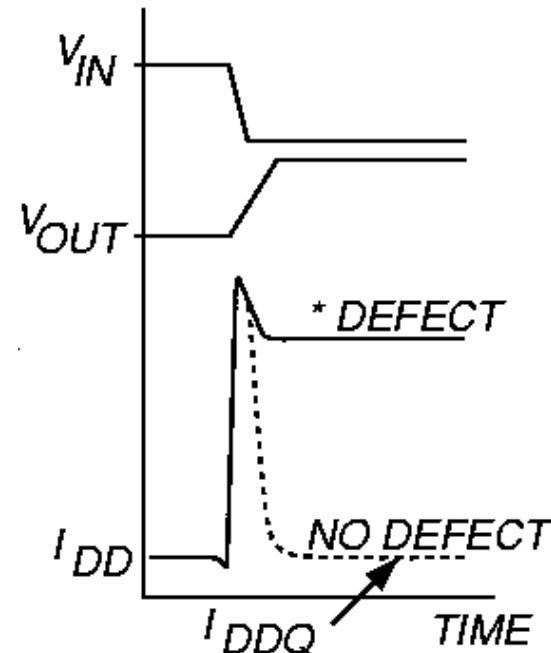
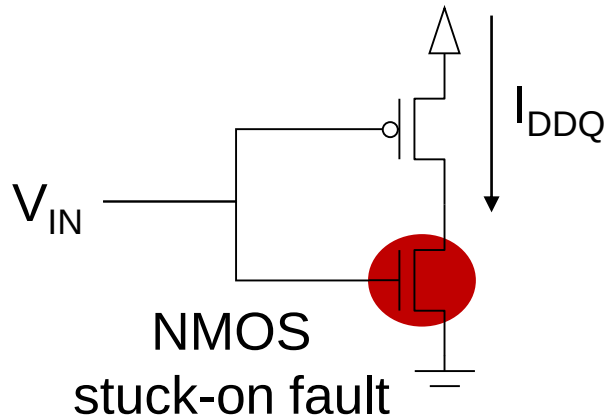
- Faulty transistor is always on
 - ♦ Form a conducting path between Vdd and Gnd in static state
- Stuck-on fault may NOT be detected by Boolean testing
 - ♦ Output logic values depends on relative impedance of transistors
- Stuck-on fault can be detected by I_{DDQ} testing
 - ♦ Apply test pattern
 - ♦ Wait for a while
 - ♦ Measure *quiescent current* from power supply



I_{DDQ} Testing [Mark 81]

- Measure static power supply current, with test patterns applied
 - ♦ I_{DD} = power supply current from V_{DD}
 - ♦ Q = quiescent
- Very commonly used in CMOS technology*
- Example: NMOS stuck-on fault

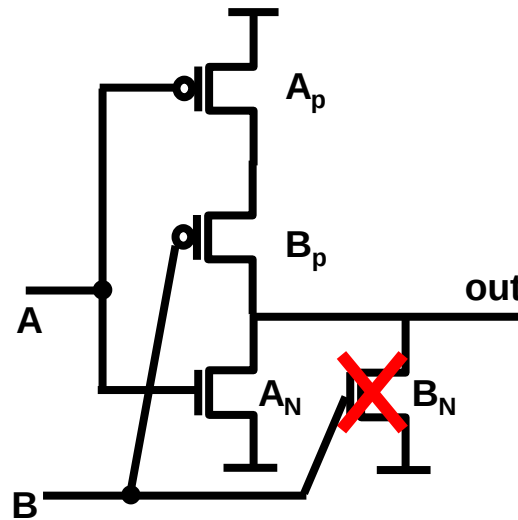
*but I_{DDQ} less effective for new technologies due to leakage



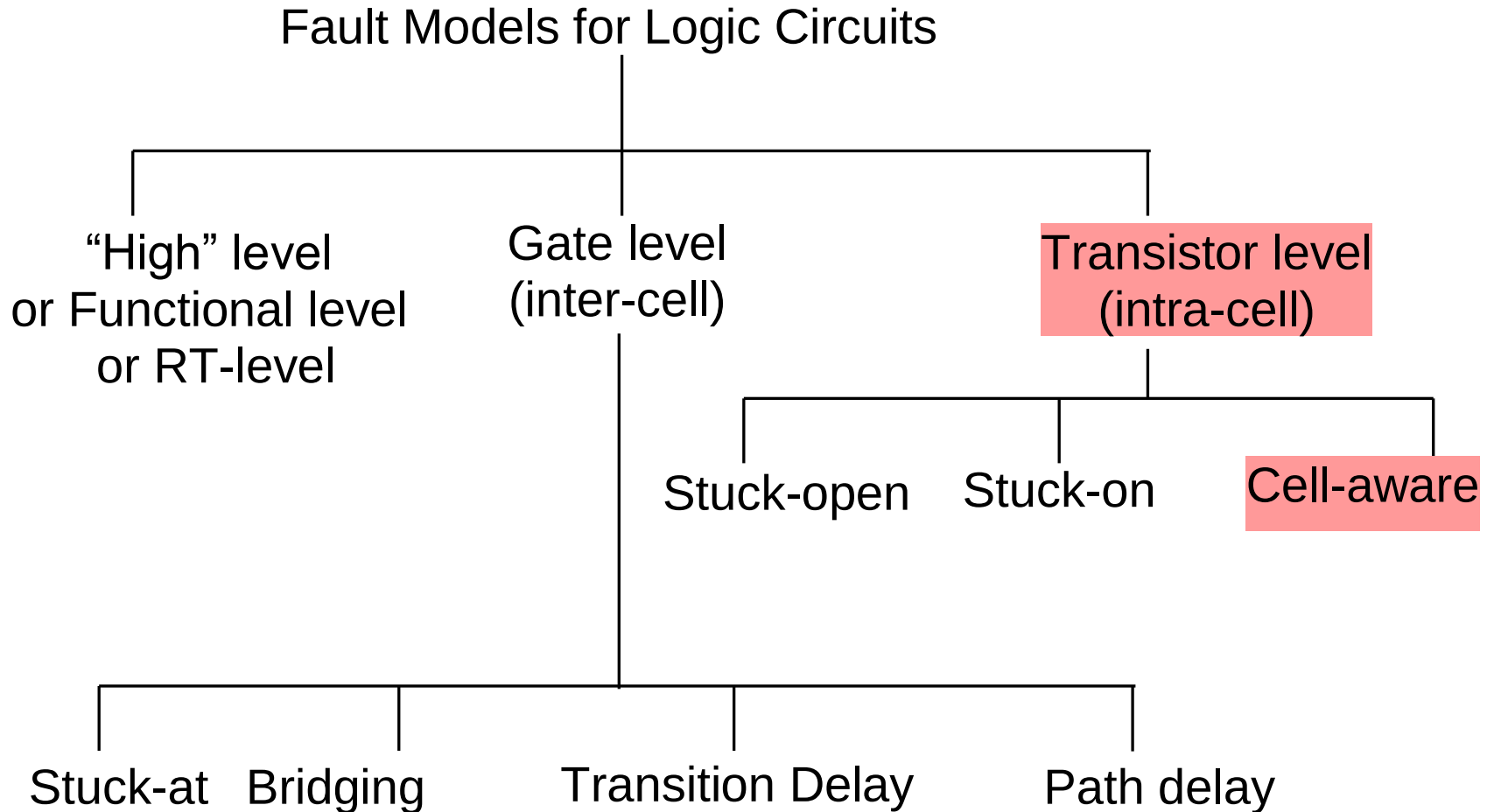
Quiz

Q: Find a test pattern to detect B_N Stuck-on fault in NOR by I_{DDQ}

A:

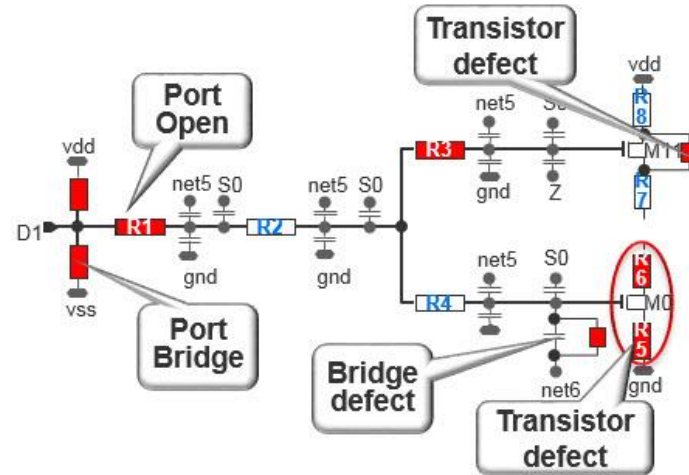
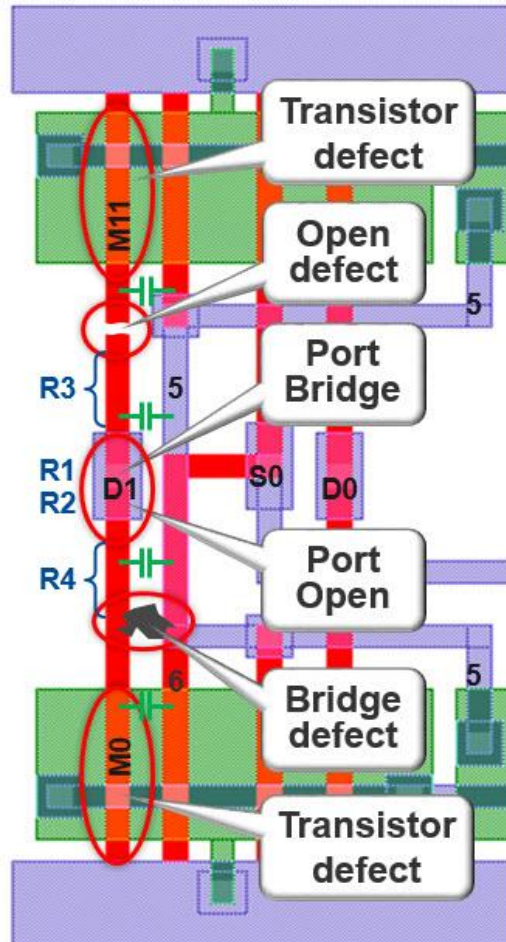


Classification of Fault Models

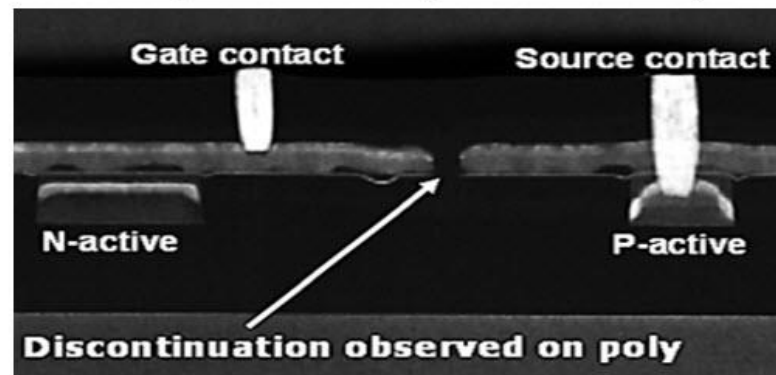


Cell-aware Test (CAT) [Hapke 09]

- Consider different defects types inside cell: open, bridge, transistor
- Need *layout extraction* and *analog fault simulation*



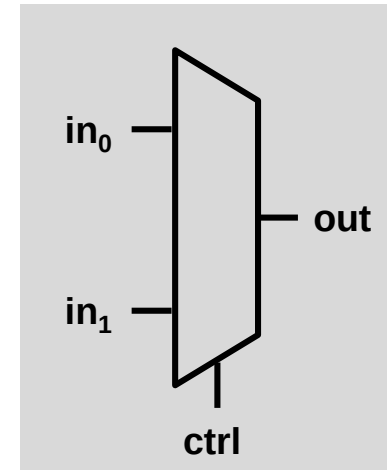
PFA proven CAT only detected defect



Why CAT more Effective?

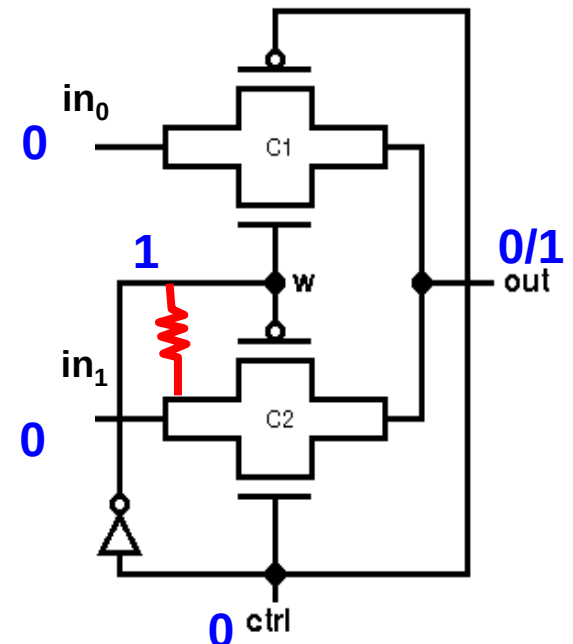
- 4 test patterns detect 8 SSF at MUX I/O pins
 - ◆ 100% SSF coverage

in_0	in_1	ctrl	out	detected SSF
0	1	0	0	ctrl SA1, out SA1, in_0 SA1
1	0	0	1	in_1 SA1
1	0	1	0	ctrl SA0, out SA0, in_0 SA0
1	1	1	1	in_1 SA0



- CAT adds {000} to detect in_1/w bridging

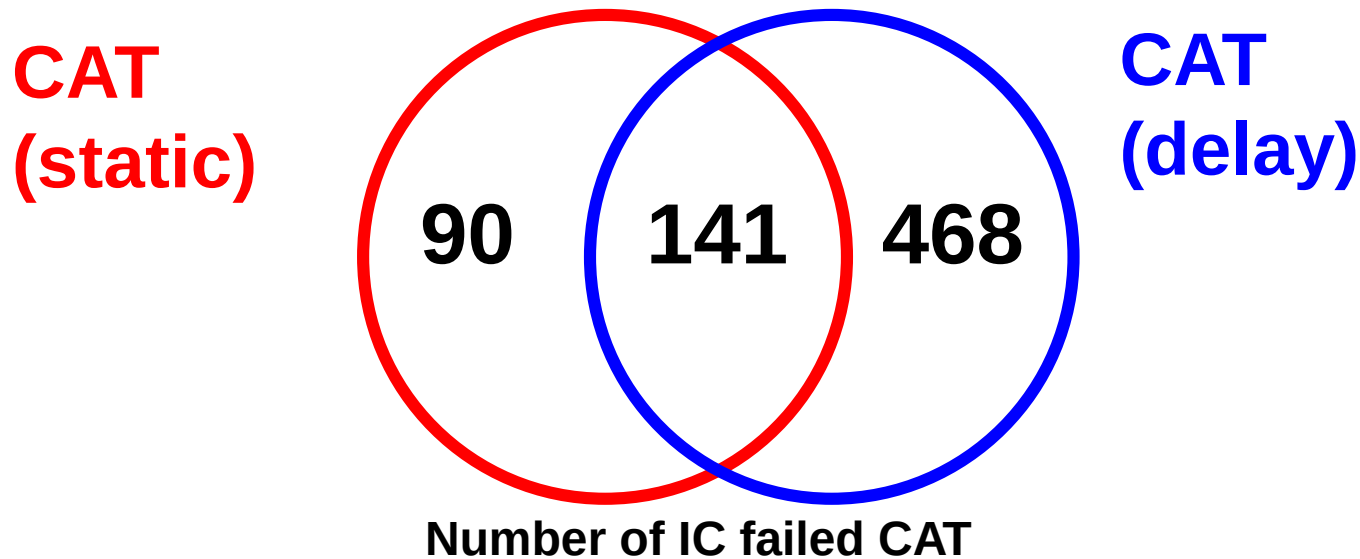
in_0	in_1	ctrl	out
0	1	0	0
1	0	0	1
1	0	1	0
1	1	1	1
0	0	0	0/1



**CAT is Effective
but Longer**

Experimental Results [Hapke 14]

- AMD 32 nm, total **800K** IC tested
- **699** IC failed only CAT, passed other tests



CAT Improves 885DPM

Fault Modeling

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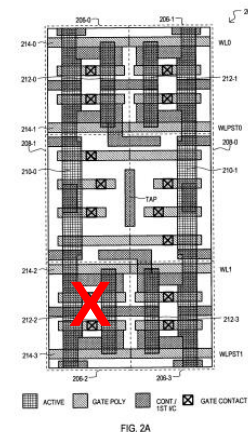
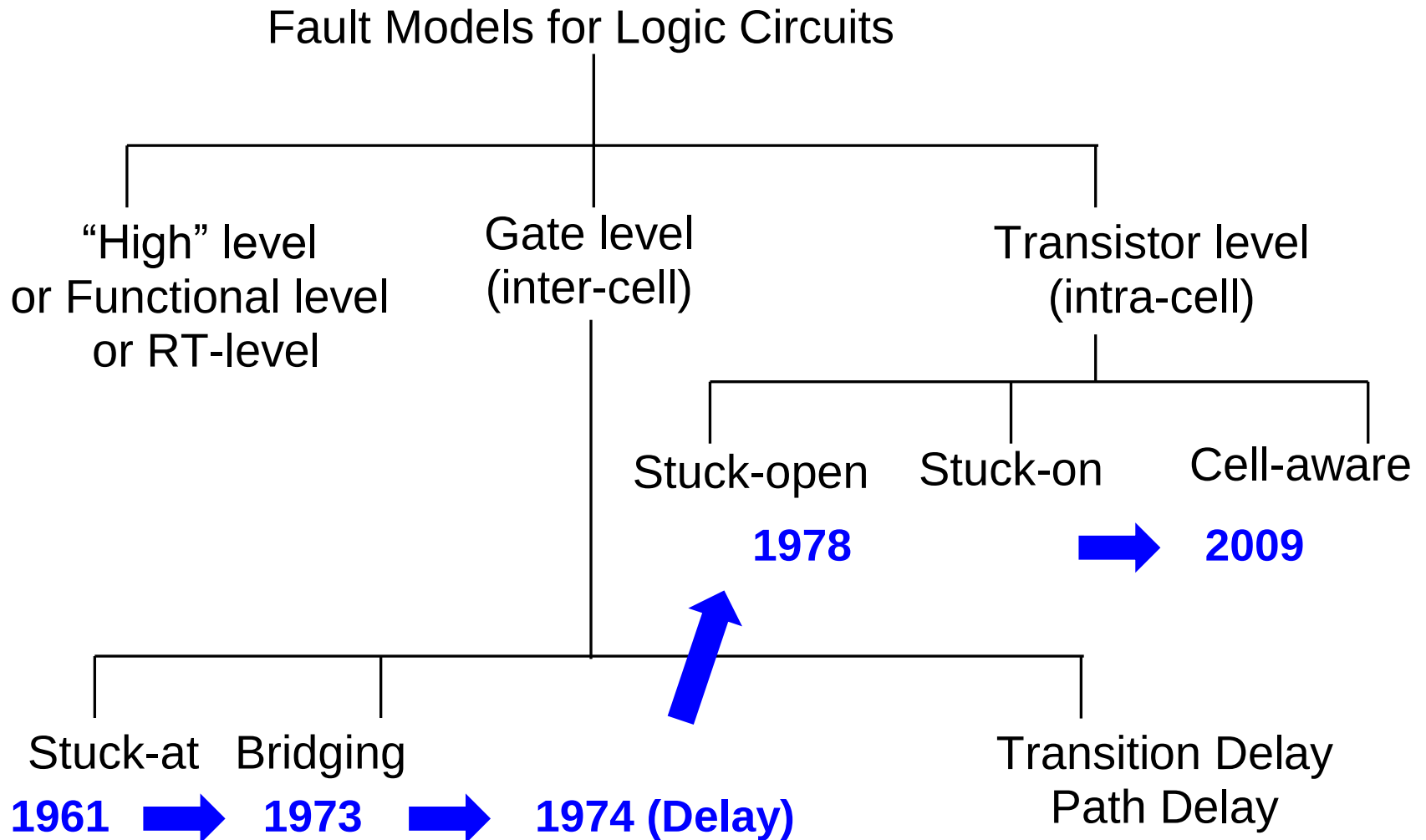


FIG. 2A

Classification of Fault Models



Fault Models Evolve with Technology

More Classifications

- According to timing
 - ◆ **AC faults**: timing dependent
 - * Delay faults
 - ◆ **DC faults**: not timing dependent
 - * Stuck-at faults, stuck-open faults ...
- According to memory effect
 - ◆ **Sequential faults**: faults that increase number of states
 - * Stuck-open faults, feedback bridging faults
 - ◆ **Combinational faults**: faults do not increase number of states
 - * Stuck-at faults, non-feedback bridging faults
- According to circuit under test (CUT)
 - ◆ **Memory** fault model (see *memory testing* chapter)
 - ◆ **FPGA** fault model
 - ◆ ...

No Single Best Fault Model for All Circuits

Summary

- ① Transistor stuck-open (SOP)
 - ◆ **Sequence dependence**: faulty comb. logic has memory
 - ◆ **TDF $\neq$ SOP**
- ② Transistor stuck-on
 - ◆ **I_{DDQ} testing**
- ③ Cell-aware testing
 - ◆ layout extraction + analog fault simulation
 - ◆ **Improved DPM** but **longer** test
- Fault models evolve with technology
 - ◆ **NO single best fault model** for all circuits
 - ◆ New fault models needed for future technologies!

